

1. Open your desktop or laptop, locate the RAM slots, and remove and re-insert the RAM module to simulate reseating — then document each step you took and any issues you observed during the process.

ANS:-

Steps followed:

1. First, I switched off the computer and disconnected the power supply.
2. I opened the cabinet and found the RAM slots on the motherboard.
3. I gently pushed the side clips to unlock the RAM module and took it out carefully.
4. I checked the module for dust or any visible damage.
5. I aligned the notch of the RAM with the slot and pushed it back until the clips were locked.
6. Finally, I closed the cabinet, switched on the PC, and checked whether it was working properly.

Observation: The RAM was fitted correctly, and the computer started without any errors. No major problem was noticed during the process.

2. Create a troubleshooting checklist for diagnosing a faulty GPU in a gaming PC, including at least 5 steps from initial symptom identification to confirming the fix. Hint: Think about display issues, driver checks, and physical connections.

ANS:-

A GPU problem can cause display errors, game crashes, or poor graphics performance. The following steps can help identify and solve the issue.

1. Identify the Problem: Check whether the screen is blank, showing unusual lines, or whether games are crashing.
2. Inspect the Connections: Make sure the graphics card is properly inserted into the motherboard and all power cables are connected.
3. Check the Monitor: Verify that the display cable is connected properly and the monitor is set to the correct input.
4. Check the Drivers: Update the graphics driver or reinstall it if it is damaged or outdated.
5. Test the GPU: Run a game or graphics test to check whether the card is performing normally.
6. Verify the Result: Restart the computer and confirm that the display and games are working without any issues.

Conclusion: These steps help identify common GPU problems and confirm whether the graphics card is functioning correctly.

3. Simulate a storage issue by unplugging and replugging a SATA or NVMe drive (SSD/HDD) on a test system, then describe the error messages or BIOS indicators you see before and after reconnecting.

ANS:-

Steps performed and observations:

1. Before Disconnecting: I checked the BIOS settings and confirmed that the SSD was detected in the storage information.
2. After Disconnecting: When the storage drive was disconnected, the BIOS could no longer detect it. The system may display an error such as "No Bootable Device" or "Boot Device Not Found."
3. Reconnecting the Drive: I switched off the system and reconnected the storage drive properly.
4. Final Check: After restarting the computer, I entered the BIOS again and verified that the SSD was visible in the storage list.
5. System Verification: The computer was able to boot normally after the drive was detected again.

Conclusion: This simulation shows that a loose or disconnected storage connection can prevent the system from starting. Properly reconnecting the drive can resolve the issue.

4. Given a scenario where a PC randomly restarts, list which component (RAM, GPU, storage) you would check first and explain your reasoning, then outline the specific diagnostic steps you would perform for that component.

ANS:-

Component to Check First: RAM

Reason: I would check the RAM first because a faulty or loose memory module can lead to unexpected restarts, system crashes, and unstable performance.

Diagnostic Steps:

1. Reseat the RAM: Turn off the PC, remove the RAM module, and insert it again properly to ensure a secure connection.
2. Test Each Module: If multiple RAM sticks are installed, check them one by one to identify whether any module is faulty.
3. Inspect the Slots: Look for dust or any visible damage in the RAM slots that could affect the connection.
4. Run a Memory Test: Use Windows Memory Diagnostic to check whether the RAM has any errors.
5. Check System Stability: Restart the computer several times and use it normally to see whether the problem occurs again.

Conclusion: If the computer works without unexpected restarts after these checks, the RAM connection or memory issue may have been the cause. If the problem continues, other components such as the GPU, storage drive, or power supply should also be checked.